Optimal Incentive Allocation Under A Fixed Budget

Trading off sample size and compliance to minimize MSE under a fixed budget

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Roadmap

1. Motivation and problem
2. Motivation
3. Setting
4. Estimator and stratum MSE bound
5. Results summary
6. Extensions

Motivation

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- ITT stays identified, but the estimand of interest is less precise.

Problem: How to buy real treatment exposure instead of just power?

What We Usually Do (and What Still Hurts)

Standard responses:

- Report ITT and live with dilution.
- Use encouragement and estimate LATE for compliers.
- Increase sample size, reminders, payments, outreach, or training.

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But each response has a cost!

- ITT targets a different estimand.
- LATE is often less precise and harder to power.
- Raising take-up or sample size is expensive.

Design question: With a fixed dollar budget, should you buy *more data* or *more compliance*?

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In this paper, we treat compliance as

1. heterogeneous across strata,
2. increasing in incentive levels, and
3. a scarce resource that can be allocated to minimize estimation uncertainty.

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- **Decision rule:** choose optimal assignment probabilities (q_k^*, r_k^*) for each stratum k to minimize risk under the budget constraint.
- **Evidence:** simulations showing when targeted designs outperform uniform assignment.

Setting

Observed per unit i:

$$\underbrace{Y_i \in \{0,1\}}_{\text{obs. outcome}}, \quad \underbrace{D_i \in \{0,1\}}_{\text{tr. assig. indicator}}, \quad \underbrace{T_i \in \{0,1\}}_{\text{tr. take-up}}, \quad \underbrace{Z_i \in \{1,2,3\}}_{\text{tr. (incentive) arms}}, \quad \underbrace{X_i = x_k}_{\text{stratum membership}}.$$

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We denote

- population weight of stratum k as (η_k)
- sample share of stratum k as (ϕ_k), and
- stratum sample size $N_k = \phi_k N$.

Our Goal

A three-arm incentive design where we choose assignment probabilities for each stratum k:

$$(q_k,\ r_k,\ 1 - q_k - r_k)$$

- q_k: assignment prob. to the **high-cost treatment arm** that **guarantees** take-up
- r_k: assignment prob. to the **low-cost treatment arm** with **partial** take-up
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Our Goal: choose (q_k, r_k) for each stratum k to minimize an MSE bound for the ATE, under a fixed dollar budget that also pins down sample size.

Three-arm Incentive Design

Within stratum k, assign:

- **Group 1 - Treatment (high-cost, guaranteed take-up):**

$$Z = 1, \quad D = 1 \text{ with prob. } (q_k), \quad T = 1$$

- **Group 2 - Treatment (low-cost, partial take-up):**

- **Group 3 - Control (low-cost, no take-up):**

Constraints:

$$q_k \geq 0, \; r_k \geq 0, \; q_k + r_k \leq 1$$

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$$Z = 2,\ D = 1 \text{ with prob. } (r_k), \text{ and } \ T = \begin{cases} 1 & \text{with prob. } \pi_{kC} \\ 0 & \text{with prob. } \pi_{kN} \end{cases} \text{ where } \pi_{kC} + \pi_{kN} = 1$$

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- **Group 3 - Control (low-cost, no take-up):**

$$Z = 3, \quad D = 0 \text{ with prob. } (1 - q_k - r_k), \text{ and } \quad T = 0$$

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Assumptions

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- **Observability:**

$$\pi_{kC} = \Pr(T = 1 \mid Z = 2, k) \text{ is identifiable from data.}$$

What must be estimated to optimize (q_k, r_k)?

Reminder: our design problem is to choose (q_k^*, r_k^*) to minimize a MSE for the ATE under a fixed budget.

To solve that problem, we first need to know:

- which stratum means are point identified from each arm,
- which components of the treated mean are not identified, and
- how that unidentified part contributes to bias and MSE.

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To solve that problem, we first need to know:

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- which components of the treated mean are not identified, and
- how that unidentified part contributes to bias and MSE.

So before optimizing the design, we derive the **estimable pieces** and **stratum-level MSE bound** to incorporate unidentified parts.

What is identified within each stratum?

Within stratum k, the design point-identifies means of two treatment arms:

$$\textbf{Arm 3 - Control:} \quad \hat{\mu}_{k0} := \frac{1}{N_k(1-q_k-r_k)} \sum_{i:Z_i=3} Y_i, \qquad \mathbb{E}[\hat{\mu}_{k0}] = \mathbb{E}[Y_k(0)].$$

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$$\textbf{Arm 1 - High Cost Tr.:} \quad \hat{\mu}_{k1} := \frac{1}{N_k q_k} \sum_{i:Z_i=1} Y_i, \qquad \mathbb{E}[\hat{\mu}_{k1}] = \mathbb{E}[Y_k(1)].$$

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And the **Arm 2 (Low-Cost Tr. arm)** identifies the treated mean **for compliers**:

$$\mathbb{E}[Y_k(1) \mid C_k] \quad \text{from units with } \{Z = 2,\ T = 1\}.$$

But **we cannot point identify not** the treated mean for **non-compliers**:

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Since that latent component lies behind the overall treated mean in **Arm 2** ($Z = 2$), we cannot point-identify it.

Next step: to derive **sharp bounds**, and then an **MSE bound**, for the estimand of interest.

A midpoint plug-in for the partially identified treated mean

Let μ_k^T denote the latent treated mean among units assigned to **Arm 2**. Then

$$\mu_{k1}^T = \pi_{kC}\mathbb{E}[Y_k(1) \mid C_k, Z_k = 2] + \pi_{kN}\mathbb{E}[Y_k(1) \mid N_k, Z_k = 2], .$$

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Problem: $\mathbb{E}[Y_k(1) \mid N_k]$ is not identified! Bounds:

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We plug in the midpoint of this interval:

$$\hat{\mu}_{k1,M} := \hat{\mu}_{k1,C}\hat{\pi}_{kC} + \frac{\hat{\pi}_{kN}}{2}.$$

This delivers a worst-case bias ($\delta_k := \frac{\pi_{kN}}{2}$) and its bounds ($\left|\mathbb{E}[\hat{\mu}_{k1,M}] - \mu_{k2}^T\right| \leq \delta_k$).

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Key tradeoff: putting more mass on $Z = 2$ can reduce variance, but it also increases exposure to bias.

A stratum-level bias-variance tradeoff

We combine:

- the **unbiased** treated mean from the guaranteed-takeup arm ($Z = 1$), and
- the **lower-variance but potentially biased** midpoint estimator from the low-incentive arm ($Z = 2$).

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Our **stratum-level ATE estimator** combines these two pieces of information:

$$\hat{\tau}_{kM}(\lambda_k) := \lambda_k \hat{\mu}_{k1,M} + (1 - \lambda_k)\hat{\mu}_{k1} - \hat{\mu}_{k0}, \qquad \lambda_k \in [0, 1].$$

Its worst-case bias is bounded by

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Under independent arms, its variance is

$$\mathrm{Var}(\hat{\tau}_{kM}(\lambda_k)) = \lambda_k^2 \mathrm{Var}(\hat{\mu}_{k1,M}) + (1 - \lambda_k)^2 \mathrm{Var}(\hat{\mu}_{k1}) + \mathrm{Var}(\hat{\mu}_{k0}).$$

So λ_k **directly controls the bias-variance tradeoff** within stratum k.

Tight stratum MSE bound and optimal λ_k^*

Within stratum k, allow arm-specific outcome variances:

$$\mathrm{Var}(\hat{\mu}_{k1}) = \sigma_{k1}^2, \quad \mathrm{Var}(\hat{\mu}_{k0}) = \sigma_{k0}^2, \quad \mathrm{Var}(\hat{\mu}_{k1,M}) = \sigma_{k1M}^2.$$

Worst-case bias magnitude from unobserved noncomplier outcomes:

$$\delta_k := \frac{\pi_{kN}}{2}.$$

With N_k units and design shares $(q_k, r_k, 1 - q_k - r_k)$,

$$\mathrm{MSE}_k(\lambda_k) \leq \lambda_k^2 \left(\frac{\sigma_{k1M}^2}{r_k N_k} + \delta_k^2 \right) + (1 - \lambda_k)^2 \frac{\sigma_{k1}^2}{q_k N_k} + \frac{\sigma_{k0}^2}{(1 - q_k - r_k) N_k}.$$

Optimal mixing weight λ_k^*

The bound is quadratic in λ_k. Minimizing over $\lambda_k \in [0,1]$ gives:

$$\lambda_k^*(q_k, r_k; N_k) = \frac{\sigma_{k1}^2/(q_k N_k)}{\sigma_{k1}^2/(q_k N_k) + \sigma_{k1M}^2/(r_k N_k) + \delta_k^2}.$$

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Interpretation:

- λ_k^* increases when the unbiased arm is noisy $\left(\sigma_{k1}^2/(q_k N_k)\right)$.
- λ_k^* decreases when the midpoint is noisy $\left(\sigma_{k1M}^2/(r_k N_k)\right)$ or when bias risk is large (δ_k^2).

Tight stratum bound at λ_k^*

Plugging λ_k^* back in yields the tight per-stratum MSE bound that we want to optimize over (q_k, r_k):

$$\overline{\mathrm{MSE}}_k^*(q_k, r_k; N_k) = \frac{\frac{\sigma_{k1}^2}{q_k N_k}\left(\frac{\sigma_{k1M}^2}{r_k N_k} + \delta_k^2\right)}{\frac{\sigma_{k1}^2}{q_k N_k} + \frac{\sigma_{k1M}^2}{r_k N_k} + \delta_k^2} + \frac{\sigma_{k0}^2}{(1 - q_k - r_k)N_k}.$$

Slack Budget Case (fixed N)

If N is fixed (or $inc_1 = inc_2$), the problem separates across strata:

$$\min_{(q_k, r_k)} \sum_{k=1}^{K} \eta_k^2 \, \overline{\mathrm{MSE}}_k^{\,*}(q_k, r_k; N_k).$$

At a fixed treated share $s_k = q_k + r_k$:

- If bias penalty δ_k^2 is large, push mass toward q_k (unbiased arm).
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Implication: with fixed N, the design problem is local to each stratum and the optimal treated split typically lies near a corner unless the bias-variance tradeoff is finely balanced.

Binding Budget Case

When the guaranteed-takeup arm is more expensive, allocating more units to it reduces the total sample size you can afford.

So each q_k has two effects:

- a **local benefit**: more guaranteed take-up improves estimation in stratum k,
- a **global cost**: the higher average cost reduces total N, which worsens precision in all strata.

Implication: under a binding budget, the problem no longer separates across strata. Choosing more guaranteed take-up in one stratum affects risk everywhere else.

Implementation Recipe

Inputs:

- Strata weights η_k and expected sample shares ϕ_k.
- Costs inc_1, inc_2 and total budget B.
- Compliance forecasts π_{kC} from pilot or historical data.
- Variance forecasts $\sigma^2_{k1}, \sigma^2_{k1M}, \sigma^2_{k0}$, or conservative upper bounds.

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Then:

1. Construct the stratum-level MSE bound for each k.
2. Map assignment choices into the feasible sample size under the budget.
3. Solve numerically for the allocation (q_k, r_k) across strata.
4. Report the optimal design, implied sample size, and resulting bound.

To Sum Up

- The three-arm design point-identifies some pieces of the treated mean and leaves one piece only partially identified.
- That missing piece can be handled with a sharp worst-case bias bound at the stratum level.
- This delivers a closed-form optimal mixing weight between the unbiased arm and the lower-variance midpoint estimator.
- The design problem then becomes how to choose (q_k, r_k) across strata to minimize risk under a fixed budget.

Practical takeaway: expensive compliance should be concentrated where it buys the largest reduction in MSE.

Simulation workflow

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3. **Estimate plug-in ingredients from the pilot.** For each stratum, estimate the bias term $\hat{\delta}_k^2$, variance components, and sample shares.

4. **Solve for the design.** Use the remaining budget to choose stratum-specific (q_k, r_k), holding the treated share fixed and accounting for how more guaranteed take-up reduces feasible sample size.

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1. **Generate a finite population.** Create $K = 3$ strata with fixed stratum weights, potential outcomes $(Y(0), Y(1))$, and compliance types that determine take-up under each arm.

2. **Run a pilot sample.** Spend a fixed share of the budget on a pilot using a common pilot allocation $(q = 0.35, r = 0.35)$ to learn about compliance and outcome variation.

3. **Estimate plug-in ingredients from the pilot.** For each stratum, estimate the bias term $\hat{\delta}_k^2$, variance components, and sample shares.

4. **Solve for the design.** Use the remaining budget to choose stratum-specific (q_k, r_k), holding the treated share fixed and accounting for how more guaranteed take-up reduces feasible sample size.

5. **Evaluate by simulation.** Under each candidate design, draw the main sample, assign treatment, estimate the midpoint and ITT estimators, and compare designs using **mean squared error across simulation replications**.

Simulation cases

What stays fixed across cases

- Stratum weights: (0.33,0.33,0.33)
- Baseline outcome levels: (0.3, 0.3, 0.3)
- Treatment effects: (0.1, 0.1, 0.1)

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- **Low:** relatively few only-high-incentive compliers (0.30, 0.10, 0.05)
- **Mid:** intermediate only-high-incentive compliance (0.6, 0.2, 0.1)
- **High:** many only-high-incentive compliers (0.90, 0.30, 0.15)

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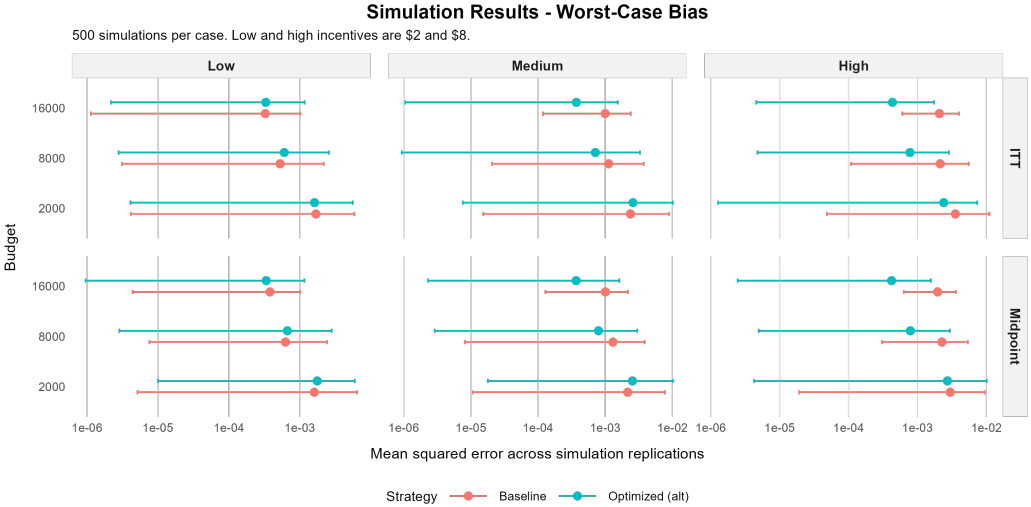
So the case variation isolates the main mechanism of interest:

when is it worth paying more for guaranteed take-up?

Reported outcome: Mean squared error (MSE) across simulation replications.

Simulation results (Worst-case bias)



Simulation results (Worst-case bias)

Case	Budget	Baseline MSE	Optimized MSE	Baseline N	Optimized N
Low	2000	0.001610	0.001770	651	504
Low	8000	0.000630	0.000670	2601	1683
Low	16000	0.000381	0.000337	5201	2987
Mid	2000	0.002170	0.002550	651	415
Mid	8000	0.001300	0.000795	2601	1368
Mid	16000	0.001010	0.000370	5201	2503
High	2000	0.002990	0.002720	651	355
High	8000	0.002270	0.000788	2601	1246
High	16000	0.001960	0.000421	5201	2371

Table: Midpoint estimator: mean MSE and average sample size by case and budget

Extension: Replacing worst-case bias with MTR

Main specification uses the worst-case bound:

$$m_{1N} \in [0, 1].$$

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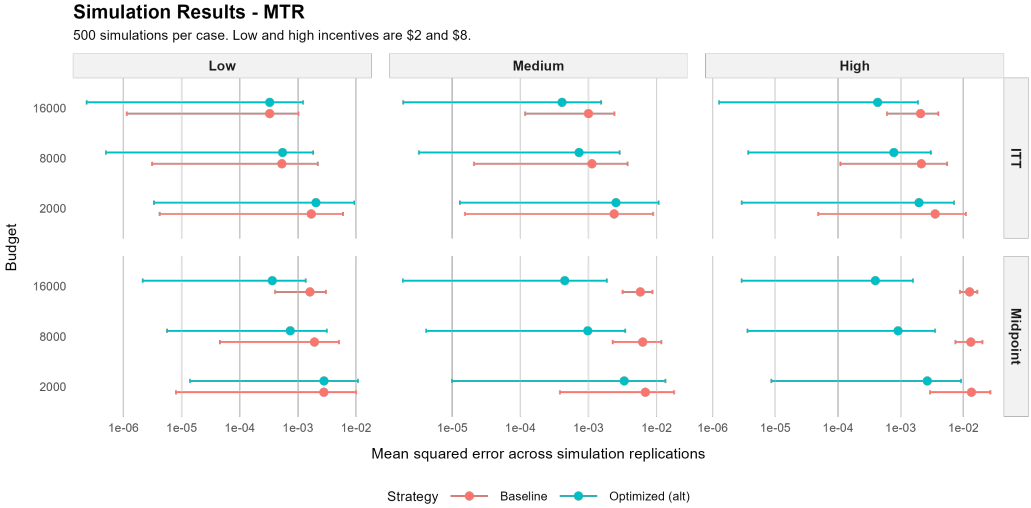
As an extension, we take advantage of the *monotone-treatment-response* (MTR) restriction:

$$m_{1N} \in [m_{0N}, 1].$$

This uses observed untreated outcomes for non-takers in $Z = 2$ to shrink the bias set, replacing the worst-case midpoint 0.5 with a tighter midpoint $(m_{0N} + 1)/2$.

Result: MTR replaces the worst-case bound with a tighter one based on observed untreated outcomes for non-takers.

Simulation results (MTR)



Simulation results (MTR)

Case	Budget	Baseline MSE	Optimized MSE	Baseline N	Optimized N
Low	2000	0.00279	0.002800	651	548
Low	8000	0.00192	0.000738	2601	1823
Low	16000	0.00161	0.000362	5201	3314
Mid	2000	0.00683	0.003350	651	465
Mid	8000	0.00625	0.000977	2601	1513
Mid	16000	0.00577	0.000449	5201	2701
High	2000	0.01350	0.002660	651	403
High	8000	0.01320	0.000910	2601	1348
High	16000	0.01260	0.000394	5201	2480

Table: Midpoint estimator: mean MSE and average sample size by case and budget

MTR with different baselines

All cases impose the same MTR assumption:

$$m_{1N} \in [m_{0N}, 1].$$

Setup:

- The same *Mid* rate of only-high incentive compliance environment.
- The only change is how untreated outcomes (p_0) line up with hard-to-induce strata (OH).

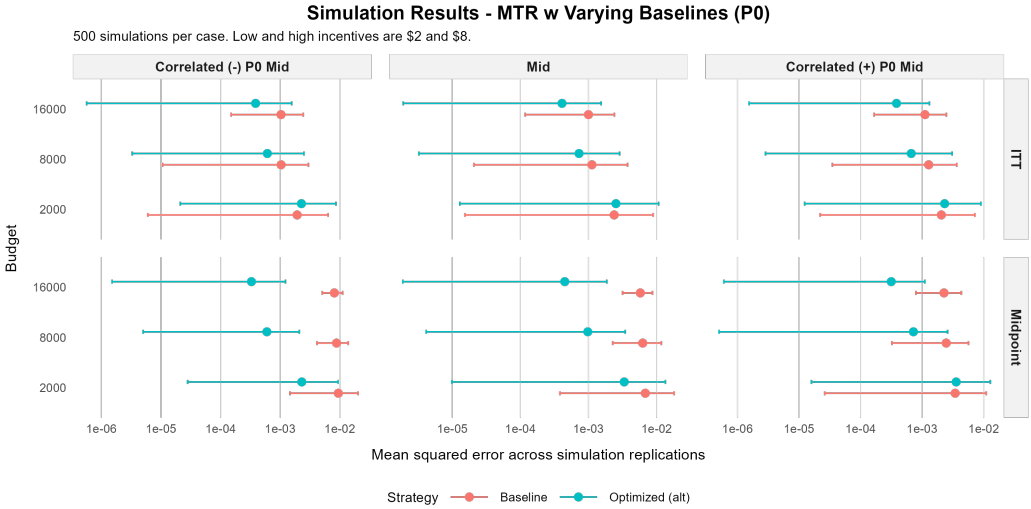
Cases

- **Fixed** p_0 as in Mid OH case before
- p_0 **positively correlated with** OH
- p_0 **negatively correlated with** OH

Expectation

- Under MTR, higher p_0 makes the midpoint arm safer.
- Gains from optimization should be **largest** under positive correlation, where MTR relaxes the problem exactly in the hardest strata.
- Gains should be **smaller** under negative correlation, where the hardest strata remain the least favorable even under MTR.

Simulation results (MTR - 2)



Simulation results (MTR - 2)

Case	Budget	Baseline MSE	Optimized MSE	Baseline N	Optimized N
Mid (MTR)	2000	0.002170	0.002550	651	415
Mid (MTR)	8000	0.001300	0.000795	2601	1368
Mid (MTR)	16000	0.001010	0.000370	5201	2503
Mid (MTR) Correlated (+) P0	2000	0.00345	0.003580	651	477
Mid (MTR) Correlated (+) P0	8000	0.00247	0.000723	2601	1427
Mid (MTR) Correlated (+) P0	16000	0.00227	0.000315	5201	2567
Mid (MTR) Correlated (-) P0	2000	0.00434	0.002650	651	499
Mid (MTR) Correlated (-) P0	8000	0.00321	0.000844	2601	1684
Mid (MTR) Correlated (-) P0	16000	0.00273	0.000446	5201	2984

Table: Midpoint estimator: mean MSE and average sample size by case and budget

Takeaway under MTR

Optimization is most valuable when the strata that are hardest to move are also the strata where MTR makes the midpoint arm safest.

That is why the gains are largest when p_0 is positively correlated with OH, and smaller when p_0 is negatively correlated with OH.

At very tight budgets, plug-in optimization can be noisy enough to slightly underperform the baseline, but the pattern becomes clear as the budget grows.

Conclusion

- Noncompliance creates a design problem: every extra unit of guaranteed take-up comes at the cost of a smaller overall sample.
- I show how to solve that problem using a stratum-specific MSE bound that combines partial identification with budget-constrained design.
- The resulting rule allocates guaranteed take-up where it buys the largest reduction in risk.
- Simulations show that uniform assignment can leave substantial efficiency gains on the table.

Takeaway: when compliance is expensive, the efficient design is to buy it selectively.

Thank you

Questions and comments welcome!

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NYU Politics Methods Workshop

References